

chip itself. This is the primary influencer for the wearables segment, with MEMS chips doubling as control units. Also, structural advances help make finer measurements of acceleration, momentum and so on," he adds.

Will we soon see the addition of communications technologies to the MEMS chip? Probably not! Prabhu explained that it is good to separate fast- and slow-moving technologies. Otherwise, our wearables would become obsolete very fast and may not support usage with a new, upgraded phone.

What is trending here

A lot of applications have long used sensors (of one sort or another) to diagnose, manage and report information. In the last few years, however, sensors have become increasingly important as devices and sub-systems have become more

Why sensor technology has evolved

Manufacturing of sensors. Miniaturisation of sensors has been possible due to advancements in material science and manufacturing technology. Improvements in material science and manufacturing technology have resulted in more reliable and predictable-quality sensors. Adoption of MEMS has increased due to advancement in process technology.

Connectivity. Wide choice of connectivity is now available based on industry application. Sensors are enabled with Ethernet, wireless mesh, Wi-Fi, ZigBee, Bluetooth and other technologies. These have been increasingly becoming smart and intelligent with integration of connectivity and processing/memory capabilities.

Processing. Processing power of sensors has increased, which has enabled these in handling more complex signal processing to digitise data. More processing can be done on the sensor or at the edge for better latency and real-time actions.

Cost and reliability. Advancement of technology has helped in bringing down the cost, triggering affordability. It has also resulted in better reliability and accuracy of measurements.

—Krishnananda Shenoy, AVP and principal architect, Engineering Services, Infosys



sophisticated with the increase in demand for reliability, especially in safety-critical applications. "The advent of real-time data capture and analysis applications is making

these new-age sensors increasingly useful in a range of applications including consumer electronics, automotive and in industrial segments," adds Jain. **EFY**

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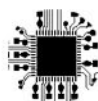
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